

PATIENT CASE

ASTar-driven treatment adjustment >66 hours earlier than standard-of-care



Name: Edward

Age: 73

A CASE OF ESCALATION

CLINICAL OVERVIEW

Edward lives in a region with a high prevalence of Antimicrobial Resistance. He has a medical history of renal failure, chronic lung disease, acute kidney injury, gastric ulcer, and myeloma.

Edward was admitted to the ICU as a follow-up to surgery after suffering from a thoracic aorta rupture.

He developed a bloodstream infection of unknown origin and symptoms of sepsis, with a **SOFA score of 6** and **SAPS score of 31**.

A blood sample was sent to the microbiology lab for pathogen ID and Antimicrobial Susceptibility Testing (AST).

Antibiotic	Breakpoints		ASTar		SoC (BMD)	
	S≤	R>	MIC	SIR	MIC	SIR
Piperacillin-Tazobactam	8	8	32	R	64	R
Meropenem	2	8	≤0.06	S	≤0.125	S

Empiric therapy

A broad-spectrum therapy of intravenous Piperacillin-Tazobactam at a dosage of 2.25 g every 8 hours was initiated while the blood sample underwent testing. His clinical condition did not improve during this time.

ASTar-driven therapy

Results from the lab identified the pathogen as *Escherichia coli* with an Enterobacterales Multi S resistance phenotype. ASTar delivered an actionable AST report to the clinical team **6 hours and 23 minutes** after the blood sample was loaded. Empiric therapy was escalated to ASTar-driven intravenous Meropenem at a dosage of 1 g every 8 hours.

Patient outcomes

Meropenem treatment was stopped after 24 days following clinical cure of the infection. Edward suffered no side effects from the antibiotics and was discharged from the hospital 32 days after starting ASTar-driven therapy.

ASTar was able to guide treatment adjustment **66.5 hours sooner than standard-of-care** (BMD) methods and **18.5 hours sooner than other routine automated AST** methods.

This patient case is a real case from clinical studies.
The name has been changed to protect patient identity.

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